Inflation, Cost of Living, and Monetary Policy

ECON 204 – Contemporary Economic Issues

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What Is Inflation?

- Inflation refers to a sustained increase in the general price level of goods and services over time.
- When inflation occurs, the purchasing power of money falls, meaning households can buy fewer goods and services with the same income.
- A one-time price increase does not necessarily constitute inflation unless it leads to continuing price growth across many sectors.
- Inflation is therefore a macroeconomic phenomenon rather than a change in the price of a single good.

The Cost of Living

- The cost of living refers to the amount of income required to maintain a given standard of living.
- Economists measure changes in the cost of living using price indices.
- The most important of these indices is the Consumer Price Index, or CPI.
- The CPI tracks the cost of a fixed basket of goods and services typically purchased by households.

How the CPI Is Constructed

- The CPI compares the cost of a representative basket of goods and services in the current period to its cost in a base year.
- The basket includes items such as food, housing, transportation, clothing, and services.
- When the CPI rises, it indicates that average consumer prices have increased.
- This increase reflects a higher cost of living for households.

Measuring Inflation

The inflation rate is calculated using the CPI formula:

$$\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100$$

- This formula measures the percentage change in consumer prices from one period to the next.
- For example, if the CPI rises from 158 to 163, inflation is approximately 3.16 percent.
- This means that average prices increased by just over three percent.

Why CPI Is Not a Perfect Measure

- The CPI may overstate inflation because it assumes households do not substitute toward cheaper goods.
- It may not fully account for improvements in product quality.
- New goods and services are not always captured immediately.
- Despite these limitations, the CPI remains the standard measure of inflation.

Demand-Pull Inflation

- Demand-pull inflation occurs when aggregate demand grows faster than productive capacity.
- Strong household spending, business investment, and government spending can increase total demand.
- When the economy is near full capacity, firms respond by raising prices rather than output.
- This leads to sustained increases in the price level.

Cost-Push Inflation

- Cost-push inflation results from rising production costs.
- Higher wages, energy prices, or supply disruptions raise firms' costs.
- Firms increase prices to maintain profitability.
- Inflation occurs alongside slower economic growth.

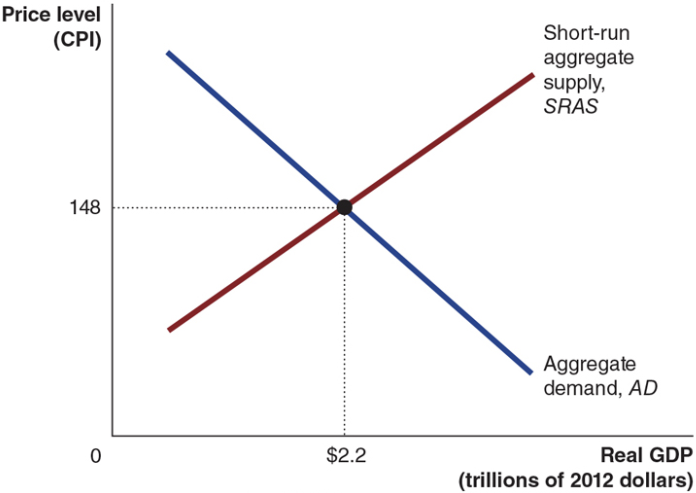
Inflation Expectations

- Inflation can persist when households and firms expect higher future inflation.
- Workers demand higher wages to protect purchasing power.
- Firms raise prices to cover higher labor costs.
- This wage-price spiral reinforces inflation over time.

Aggregate Demand and Supply

- Aggregate demand represents total spending in the economy.
- Aggregate supply represents total output firms are willing to produce.
- The intersection of AD and AS determines the price level and real GDP.

AD–AS Diagrams



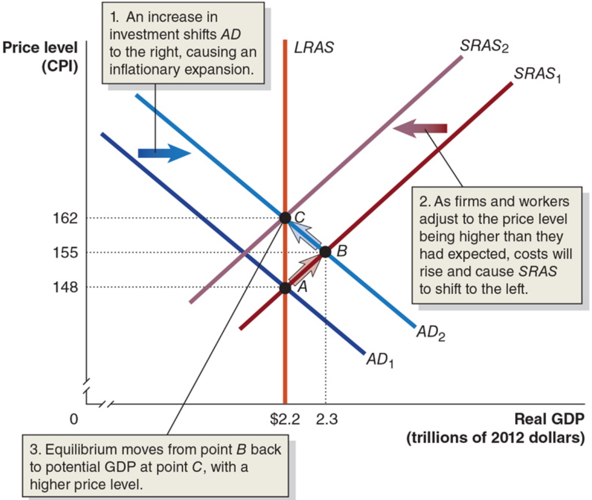
Demand-Pull Inflation in AD–AS

- An increase in aggregate demand shifts the AD curve to the right.
- In the short run, both output and prices rise.
- Over time, rising costs shift short-run aggregate supply leftward.
- Output returns to potential while the price level remains higher.

Demand-Pull Inflation

Figure 9.6 The Short-Run and Long-Run Effects of an Increase in Aggregate Demand

In the short run, an increase in aggregate demand causes an increase in real GDP. In the long run, it causes only an increase in the price level.



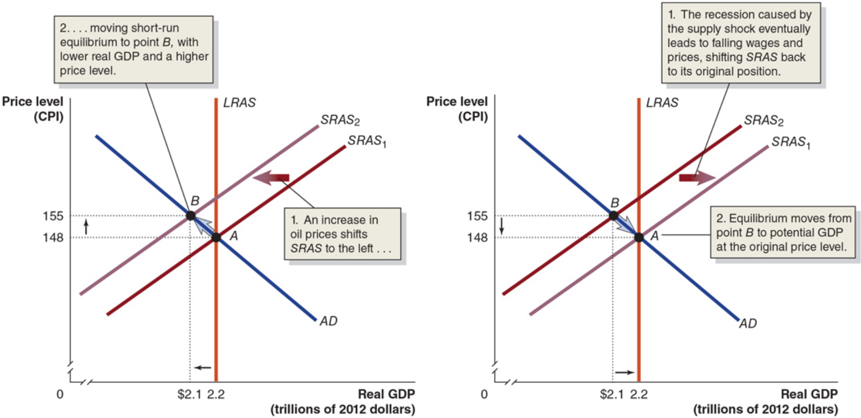
Cost-Push Inflation in AD–AS

- Rising production costs shift short-run aggregate supply leftward.
- Prices rise while output falls.
- Policymakers face a trade-off between inflation control and growth.

Cost-Push Inflation

Figure 9.7 The Short-Run and Long-Run Effects of a Supply Shock

Panel (a) shows that a supply shock, such as a large increase in oil prices, will cause a recession and a higher price level in the short run. The recession caused by the supply shock increases unemployment and reduces output. In panel (b), rising unemployment and falling output result in workers being willing to accept lower wages and firms being willing to accept lower prices. The short-run aggregate supply curve shifts from $SRAS_2$ to $SRAS_1$. Equilibrium moves from point *B* back to potential GDP and the original price level at point *A*.



(a) A recession with a rising price level—the short-run effect of a supply shock

(b) Adjustment back to potential GDP—the long-run effect of a supply shock

Economic Costs of Inflation

- High inflation creates uncertainty for households and firms.
- Planning long-term contracts becomes more difficult.
- Inflation redistributes income in arbitrary ways.
- Fixed-income households may lose purchasing power.

Borrowers and Lenders

- Unexpected inflation benefits borrowers.
- Loan repayments are made in money with lower purchasing power.
- Lenders receive less real value than expected.

Nominal vs Real Values

- Nominal values are measured in current dollars.
- Real values are adjusted for inflation.
- A nominal wage increase does not guarantee higher purchasing power.

Real Interest Rates

$$r \approx i - \pi^e$$

- The real interest rate reflects the true cost of borrowing.
- It depends on expected inflation.
- It influences consumption, saving, and investment decisions.

The Bank of Canada

- Monetary policy in Canada is conducted by the Bank of Canada.
- The main objective is price stability.
- Inflation targeting provides a clear policy framework.

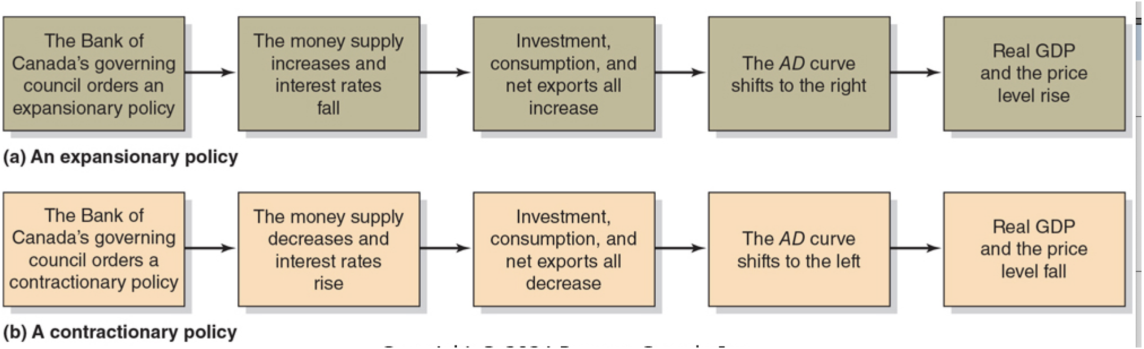
Policy Interest Rate

- The Bank sets a target for the overnight interest rate.
- This influences borrowing costs throughout the economy.
- Commercial banks adjust their lending rates accordingly.

Transmission Mechanism

- Higher rates reduce household consumption.
- Business investment declines.
- Housing demand slows.
- The exchange rate may appreciate.

Monetary Policy



Why Expectations Matter

- Stable inflation expectations support price stability.
- Firms and workers adjust prices and wages accordingly.
- Policy credibility reduces the need for aggressive rate changes.

Key Takeaways

- Inflation reduces purchasing power over time.
- The CPI measures changes in the cost of living.
- Monetary policy influences inflation through interest rates.
- Expectations play a crucial role in stabilizing the economy.